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Assignment 4

Question 1: Ways to advise against cybercrime and a framework to improve protection

Cybercrime means crimes done using computers or the internet, such as hacking, phishing, fraud and stealing data. For an organization, the best way to advise against it is to use a security framework. A framework is a set of steps and rules that helps the organization protect itself in an organized way. For this assignment I chose the NIST Cybersecurity Framework, which has six functions.

- Govern: The organization makes a security policy and decides who is responsible. It must also follow the laws on cybercrime and data protection.
- Identify: The organization lists its assets like computers, data and software, and finds out what risks they face.
- Protect: Use strong passwords, multi-factor authentication, encryption and regular updates. Staff are also trained so they do not fall for phishing emails.
- Detect: Use monitoring tools to notice strange activity early, like many failed logins.
- Respond: Have an incident response plan so everyone knows what to do when an attack happens.
- Recover: Keep backups and practice restoring systems so the business can continue after an attack.

Other ways to advise against cybercrime are having an acceptable use policy that tells workers that hacking and misuse of data are punished by the company and by law, doing regular awareness training, and giving staff a safe way to report suspicious activity. Using a framework like this makes the organization safer and shows that it takes security seriously.

Question 2: Role of risk management software

Risk management software helps an organization find, measure and control risks in one place. Instead of using many separate spreadsheets, the organization can manage all risks using one system. Examples are RSA Archer and ServiceNow. Its main roles are:

- Keeps a risk register that lists each risk, who owns it, how likely it is and how serious it would be.
- Scores the risks so the organization knows which ones to deal with first.
- Helps check that the organization follows standards and laws like ISO 27001 and GDPR.
- Connects with security tools such as vulnerability scanners to give updated information.
- Gives tasks to people to fix risks and tracks until they are finished.
- Makes reports and dashboards so managers can make good decisions.

In conclusion, risk management software saves time, reduces mistakes and helps the organization manage risks all the time and not only once in a while.

Question 3: Scenario for stakeholders on security issues across platforms

Scenario: An online shop is about to launch a new platform. It will have a website, a mobile app for Android and iPhone, and a link to another company's marketplace. Customers will save their card details and addresses. Staff will also use their personal phones to access the admin system. Before launching, the security team wants to explain the risks to the stakeholders.

- Website: Attackers can use SQL injection or cross-site scripting to steal customer information.
- Mobile app: Data saved on the phone may not be safe, and the app may run on phones that are rooted or not updated.
- Third party links: If the partner company is hacked or a key is leaked, our system can also be affected.
- Cloud: A wrong setting can make stored data public by mistake.
- Staff phones: Phishing or malware can steal staff passwords.

If these problems happen, the company can lose money, pay fines, lose customers' trust, and have its system stop working. For this reason, the stakeholders are asked to approve money for multi-factor authentication, a web application firewall, security testing of the apps, staff training and a monitoring service before the launch.

Question 4: Database security service for information generated on the platform

The platform creates a lot of information such as customer details, orders and payment data. This data must be protected from hackers, workers who misuse it, and accidental loss. The database security service I propose has these parts:

- Access control: Only people who need the data can get it (least privilege). Admins must use multi-factor authentication, and each user has their own account.
- Encryption: Data is encrypted when it is moved (TLS) and when it is stored (AES-256). The encryption keys are kept in a separate safe place.

- Data protection: Card and personal details are masked or tokenized, and data is grouped by how sensitive it is.
- Safe coding: The application uses parameterized queries and input checks to stop SQL injection.
- Monitoring: The database activity is logged and checked. Alerts are sent when there are strange actions like downloading a large amount of data.
- Backups: Backups are encrypted and made regularly, and restoring is tested so data can be recovered.
- Network security and updates: The database is kept in a private network, not open to the internet, and is patched and scanned for weaknesses regularly.
- Rules: There are policies on how long data is kept and how it is deleted, to follow data protection laws.

These layers together protect confidentiality, integrity and availability of the data. If one layer fails, the others still give protection.

Conclusion

Cybercrime is a serious threat to every organization. Using a framework, risk management software, clear communication with stakeholders and a strong database security service helps the organization reduce risks and protect its data and its customers.